A train of gravitational waves sweeps into the solar system from a distant, cosmic source. Each wave crest hits the Japanese detectors first, then sweeps through the Earth to the Washington detectors, then Louisiana, and finally Italy. For roughly a minute, crest is followed by trough is followed by crest. The masses in each detector wiggle ever so slightly, perturbing their laser beams and hence perturbing the light that enters the detector's photodiode. The eight photodiode outputs are transmitted by satellite links to a central computer, which alerts a team of scientists that another minute-long gravitational-wave burst has arrived at Earth, the third one this week. The computer combines the eight detectors' outputs to produce four things: a best-guess location for the burst's source on the sky; an error box for that best-guess location; and two *waveforms*—two oscillating curves, analogous to the oscillating curve that you obtain if you examine the sounds of a symphony on an oscilloscope. The history of the source is encoded in these waveforms (Figure 10.9).

There are two waveforms because a gravitational wave has two *polarizations.* If the wave travels vertically through an interferometer, one polarization describes tidal forces that oscillate along the east—west and north-south directions; the other describes tidal forces oscillating along the northeast—southwest and northwest—southeast directions. Each detector, with its own orientation, feels some combination of these two polarizations; and from the eight detector outputs, the computer reconstructs the two waveforms.

The computer then compares the waveforms with those in a large catalog, much as a bird watcher identifies a bird by comparing it with pictures in a book. The catalog has been produced by simulations of sources on computers, and by five years of previous experience monitoring gravitational waves from colliding and coalescing black holes, colliding and coalescing neutron stars, spinning neutron stars (pulsars), and supernova explosions. The identification of this burst is easy (some others, for example, from supernovae, are far harder). The waveforms show the unmistakable, unique signature of two black holes coalescing.